

SunMint Program — Consolidated Progress Report (v5)

Prepared by: Sophia Truesight (autopilot) · **For:** Gary Teh & TrueSight DAO members · **Date:** 2026-08-26

Revision note: v5 adds **Section 5 — FREE SATELLITE ACCESS:** the APIs that make "free satellite" actually real (with commercial-use terms), plus everything from v4 (PM002 formula, lean stack, resource gaps).

1. The model — mobile-first, decentralized

SunMint Farmer App (Android + iOS) — `TrueSightDAO/sunmint\_mobile`, Capacitor 8, 21+ PRs merged, UAT'd 2026-08-23. Offline-first, RSA-signed, PT/EN. The phone is the farmer's carbon-MRV device.

2. Certification route (decided)

- **Pilot → Plan Vivo (PVCs) under PM002** — the Acorn methodology (active 29/09/2025): participatory farmer monitoring, lower validation costs, ≥60% of revenue to communities
- **Scale → Verra VM0047 (ARR) + CCB** — ICVCM-approved; corporate offtake path
- VM0017 corrected (SALM ≠ ARR) — PR #294

3. Landmark precedents

- **Andean Cacao (Colombia)** — first large-scale cacao agroforestry under Verra: VM0047 + CCB, Terra Global, 56,000+ VCUs first issuance
- **ACORN (Rabobank + Solidaridad + Microsoft + Satelligence)** — cocoa/coffee agroforestry, Plan Vivo PM002, ~80% of credit value to farmers, 4–6 tCO₂e/ha/yr
- **TREEO** — calibration-card DBH via phone camera, 94–95% accuracy ($R^2 \geq 0.95$)
- **Greenstand Treetracker** — geotagged repeat photos, 500k+ trees, open-source
- **CommuniTree (Taking Root)** — 28M trees, Plan Vivo since 2010

4. THE METHODOLOGY — ACORN / Plan Vivo PM002 (the actual formula)

Source: Plan Vivo Climate Methodology PM002 v1.0 (developer: Acorn; reviewers: Plan Vivo TRP & AENOR; active 29/09/2025).

Step 1 — Belowground biomass (Equation 1):

...

$\Delta\text{BGB}_p = \Delta\text{AGB}_p \times R$ R = root:shoot ratio (IPCC 2019 Table 4.4; default 0.32)

...

Step 2 — Plan Vivo Certificates (Equations 6.1 & 6.2):

...

$$\text{rPVCs} / \text{vPVCs} = ((\Delta\text{AGB}_p + \Delta\text{BGB}_p) \times 0.47) \times (44/12)$$

$$\times (1 - A_{\text{pre}}) \times (1 - A_{\text{unc}}) \times (1 - \text{LD_CE})$$

$$\times (1 - \text{AR}) \times (1 - \text{RB}) - (E_{\text{proj},p} - E_{\text{base},p})$$

...

- CF = carbon fraction = **0.47** · F = C→CO₂ = **44/12 = 3.667**
- AR = achievement reserve = **10%** · RB = risk buffer contribution = **20%**
- A_{pre} (pre-project biomass), A_{unc} (uncertainty), LD_CE (leakage) per modules PU007/PU008/PU004

Step 0 — the tree measurement (phone, PU006):

- DBH at 1.3 m via calibration-card photo (TREEO) → species allometric equation → AGB (kg)
- SunMint needs equations for **cacao, Brazil nut, açai, mahogany, jatobá** (peer-reviewed or locally calibrated)

The formula is fully implementable in our mobile app — per-tree DBH → AGB → ΔAGB → Eq 6.1/6.2 → PVCs.

5. ■ FREE SATELLITE ACCESS — the APIs (what makes "\$0 satellite" real)

Yes — there are free APIs. The satellite layer costs nothing. Three production-grade, free, commercial-use-licensed APIs:

5.1 Copernicus Data Space Ecosystem (ESA) — Sentinel

- **What:** Sentinel-2 (10 m optical, 5-day revisit), Sentinel-1 (radar), Sentinel-3
- **Access:** Copernicus Browser (web) + **OData API** + cURL/python (STAC + OData endpoints)
- **License:** free, full and open data policy — **commercial use explicitly allowed** (attribution: "credit Copernicus appropriately"); free account with fair-use quota
- **Docs:** `dataspace.copernicus.eu` — search via OData/STAC, download GeoTIFF directly
- **Use for:** baseline (non-forest ≥10 yr proof), annual NDVI/land-cover, leakage context

5.2 Microsoft Planetary Computer — Sentinel-2 + Landsat + HLS

- **What:** petabyte archive: Sentinel-2-L2A, Landsat 8/9, HLS (harmonized), NAIP, more — all in one **STAC API**
- **Access:** free public STAC API (`planetarycomputer.microsoft.com/api/stac/v1`) + auth API for signed asset URLs; Python SDK (`pystac-client`, `planetary-computer`)
- **License:** **free for commercial use** (public catalog, not Pro)
- **Use for:** wall-to-wall time series, NDVI/GBI vegetation indices, plot-scale extraction via GeoJSON

5.3 NASA Earthdata / USGS EarthExplorer — Landsat

- **What:** Landsat 8/9 (30 m, 16-day), full 50-year archive
- **Access:** USGS EarthExplorer + NASA Earthdata Search; AppEEARS for subset extraction
- **License:** public domain, free for any use (USGS policy)
- **Use for:** the deepest historical record — proving long-term non-forest land use for the VM0047 baseline

Comparison for SunMint

API	Res.	Revisit	Baseline depth	Best for
Copernicus CDSE (Sentinel-2)	10 m	5-day	2015→	NDVI growth curves, plot cover
MS Planetary Computer (Sentinel-2/ Landsat/HLS)	10–30 m	5–16-day	1984→	Time series + analysis-ready data
USGS EarthExplorer (Landsat)	30 m	16-day	1972 →	VM0047 non-forest baseline proof

Bottom line: the baseline proof and annual monitoring can be built on **free APIs** — no NOR Space subscription required at launch. NOR becomes purely optional optimization. (VM0047 caveat unchanged: >50 planting units/ha → area-based approach → remote-sensing baseline is mandatory, but the *free* tier satisfies it.)

6. ■■■ RESOURCE GAPS — what we need to execute (explicit)

People / capability

Gap	Need	Est. cost
PDD/PM002 document writer	Carbon methodology consultant (agroforestry)	\$8–15k
VVB engagement	Plan Vivo TRP-approved body (AENOR, Earthood)	\$10–20k
Field enumerator training	Calibration-card method training	\$3–6k
Allometric equations	5 species (cacao, Brazil nut, açai, mahogany, jatobá)	\$5–10k

Data (satellite part now FREE — only GIS labor needed)

Gap	Need	Source
Baseline land-use proof	Non-forest ≥10 yr evidence	FREE: Copernicus CDSE / USGS Landsat archive (GIS analyst ~2–4 wks)

Gap	Need	Source
Plot boundaries (GIS)	Parcel polygons per co-op	Survey + CAR records
CAR tenure docs	Land registration numbers	Co-ops
Community engagement records	FPIC / meeting minutes	Co-ops + us

Technology

Gap	Need	Effort
`monitor_tree_growth` module	Calibration-card photo → DBH → allometric CO ₂ e → signed event	2–3 wks
Tree registry	Per-tree IDs in ledger	1–2 wks
Allometric engine	Species formula table (5 species)	1 wk
Satellite baseline pipeline	STAC/OData ingestion (free APIs above)	2–4 wks

Financial (lean stack)

Item	Est.
Stage 0 — legal/tenure + baseline GIS (free satellite) + community records	\$5–10k
Stage 1 — PDD + VVB + calibration	\$35–60k
Stage 2 — pilot planting 20–50 ha	\$150–250k
Stage 3 — scale to 10,000 ha	\$3–10M+

The single biggest gap

Baseline data + PDD validation — with the free satellite APIs, the baseline proof is now a GIS-labor cost (not a satellite-subscription cost). Nothing else moves until the PDD is written + VVB-attested.

7. Prioritized execution path (lean stack)

1. **Merge PR (lean stack + PM002)** — done this cycle
2. **Stage 0 (~\$5–10k DAO injection):** legal/tenure, free-satellite baseline GIS, community records
3. **Build `monitor\_tree\_growth` module** (implements PM002 equation chain)
4. **Shortlist VVBs** (AENOR/Earthood — the actual PM002 reviewers)
5. **Develop local allometric equations** for the 5 species
6. **Submit Terra RFP** once baseline + boundaries + tenure documented
7. **Optional:** NOR Space only if VVB demands premium analysis; PODream drone calibration post-pilot

Prepared by Sophia Truesight, autopilot · TrueSight DAO · Mission: restore 10,000 hectares of Amazon rainforest